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Semiconductor device and method of fabricating the same

Abstract

A semiconductor device includes a lower structure and an upper structure on the lower structure. The lower structure includes a first semiconductor substrate, a first pad and a first dielectric layer. The first dielectric layer surrounds the first pad and exposes a top surface of the first pad. The upper structure includes a second semiconductor substrate, a second pad and a second dielectric layer. The second dielectric layer surrounds the second pad and exposes a bottom surface of the second pad. The first pad and the second pad are bonded to each other across an interfacial layer to couple the upper and lower structures to each other. The first and second pads and the interfacial layer include a same metallic material. The first and second pads have a substantially same average grain size and the interfacial layer has a different average grain size than the first and second pads.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority under 35 U.S.C § 119 to Korean Patent Application No. 10-2022-0090194, filed on Jul. 21, 2022 in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference in its entirety herein.

TECHNICAL FIELD

(2) The present disclosure relates generally to a directly bonded semiconductor device and a method of fabricating the same.

DISCUSSION OF RELATED ART

(3) In the semiconductor industry, consumer demand for high capacity, thin, and small/miniatuized semiconductor devices and electronic products using the same has increased. Various package techniques are in development to provide semiconductor devices having these features. One approach of the various package techniques is a packaging technique which vertically stacks a plurality of semiconductor chips to achieve a high density chip stacking. This packaging technique has an advantage of integrating semiconductor chips having various functions on a relatively small area as compared to a conventional package consisting of one semiconductor chip.

(4) A semiconductor package is provided to implement an integrated circuit chip that is applied to electronic products. A semiconductor package is typically configured such that a semiconductor chip is mounted on a printed circuit board and bonding wires or bumps are used to electrically connect the semiconductor chip to the printed circuit board. With the development of electronic industry, research is being conducted for semiconductor packages having increased reliability and durability.

SUMMARY

(5) Some embodiments of the present inventive concept provide a semiconductor device with improved structural stability and a method of fabricating the same.

(6) Some embodiments of the present inventive concept provide a semiconductor package with increased electrical properties and increased operating stability and a method of manufacturing the same.

(7) Some embodiments of the present inventive concepts provide a semiconductor fabrication method in which the occurrence of failure is reduced and a semiconductor device fabricated by the same.

(8) According to some embodiments of the present inventive concepts, a semiconductor device includes a lower structure. An upper structure is on the lower structure. The lower structure includes a first semiconductor substrate. A first pad is on the first semiconductor substrate. A first dielectric layer is on the first semiconductor substrate. The first dielectric layer surrounds the first pad and exposes a top surface of the first pad. The upper structure includes a second semiconductor substrate. A second pad is on the second semiconductor substrate. A second dielectric layer is on the second semiconductor substrate. The second dielectric layer surrounds the second pad and exposes a bottom surface of the second pad. The first pad and the second pad are bonded to each other across an interfacial layer to couple the upper structure and the lower structure to each other. The first pad, the second pad, and the interfacial layer include a same metallic material. The first and second pads have a substantially same average grain size as each other and the interfacial layer has a different average grain size than the first and second pads.

(9) According to some embodiments of the present inventive concepts, a semiconductor device includes a lower structure. An upper structure is on the lower structure. The lower structure includes a first semiconductor substrate. A first pad is on the first semiconductor substrate. A first dielectric layer is on the first semiconductor substrate. The first dielectric layer surrounds the first pad and exposes a top surface of the first pad. The upper structure includes a second semiconductor substrate. A second pad is on the second semiconductor substrate. A second dielectric layer is on the second semiconductor substrate. The second dielectric layer surrounds the second pad and

exposes a bottom surface of the second pad. The first pad and the second pad are bonded to each other across an interfacial layer to couple the upper structure and the lower structure to each other. An interface between the first dielectric layer and the second dielectric layer is positioned at a level between a top surface and a bottom surface of the interfacial layer.

(10) According to some embodiments of the present inventive concepts, a method of fabricating a semiconductor device includes forming an upper structure and a lower structure. The forming of the upper and lower structures includes forming a dielectric layer on a semiconductor substrate. A hole is formed in the dielectric layer. A plating process is performed. The plating process forms a conductive layer that fills the hole and covers the dielectric layer. A planarization process is performed on the conductive layer. The planarization process forms a pad in the hole. The pad includes a recess positioned on a top surface of the pad and extending towards the semiconductor substrate. A deposition process is performed. The deposition process forms a surface layer on the top surface of the pad. An argon (Ar) plasma treatment process is performed on the surface layer and the dielectric layer. After performing the argon plasma treatment process, a nitrogen (N) plasma treatment process is performed on the surface layer and the dielectric layer. The upper structure and the lower structure are placed in direct contact with each other. The placing vertically aligns the upper structure and the lower structure with each other and a dielectric layer of the upper structure directly contacts a dielectric layer of the lower structure. An annealing process is performed on the upper structure and the lower structure. After the nitrogen plasma treatment process is performed, copper nitride (Cu₃N₄) is formed on a top surface of the surface layer and silicon hydroxide (Si—OH) is formed on a top surface of the dielectric layer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates a cross-sectional view showing a semiconductor device according to some embodiments of the present inventive concept.
- (2) FIG. 2 illustrates an enlarged view showing section A of FIG. 1 according to some embodiments of the present inventive concept.
- (3) FIG. 3 illustrates an enlarged view showing section B of FIG. 2 according to some embodiments of the present inventive concept.
- (4) FIGS. 4, 5, 6 and 7 illustrate cross-sectional views of Section A depicted in FIG. 1, showing a semiconductor device according to some embodiments of the present inventive concept.
- (5) FIG. 8 illustrates a cross-sectional view showing a semiconductor device according to some embodiments of the present inventive concept.
- (6) FIG. 9 illustrates a plan view showing a semiconductor device according to some embodiments of the present inventive concept.
- (7) FIG. 10 illustrates a cross-sectional view showing a semiconductor device according to some embodiments of the present inventive concept.
- (8) FIGS. 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25 and 26 illustrate cross-sectional views showing a method of fabricating a semiconductor device according to some embodiments of the present inventive concept.

DETAILED DESCRIPTION OF EMBODIMENTS

- (9) The following will now describe a semiconductor device according to the present inventive concept with reference to accompanying drawings.
- (10) FIG. 1 illustrates a cross-sectional view showing a semiconductor device according to some embodiments of the present inventive concept. FIG. 2 illustrates an enlarged view showing section A of FIG. 1. FIG. 3 illustrates an enlarged view showing section B of FIG. 2.
- (11) Referring to FIG. 1, a semiconductor device may include a lower structure 10 and an upper

structure **30** stacked on the lower structure **10**.

(12) The lower structure **10** may include a first substrate **12**, a first circuit layer **14**, a first dielectric layer **16**, and first pads **20**.

(13) In an embodiment, the first substrate **12** may be a semiconductor substrate, such as a semiconductor wafer. The first substrate **12** may be a bulk silicon substrate, a silicon on insulator (SOI) substrate, a germanium substrate, a germanium on insulator (GOI) substrate, a silicon-germanium (SiGe) substrate, or an epitaxial film substrate obtained by performing a selective epitaxial growth (SEG) process. The first substrate **12** may include, for example, at least one compound selected from silicon (Si), germanium (Ge), silicon germanium (SiGe), gallium arsenic (GaAs), indium gallium arsenic (InGaAs), aluminum gallium arsenic (AlGaAs), and a mixture thereof. Alternatively, the first substrate **12** may be a dielectric substrate, such as a printed circuit board (PCB).

(14) The first circuit layer **14** may be disposed on the first substrate **12**. The first circuit layer **14** may be disposed on a top surface of the first substrate **12**. The first circuit layer **14** may include a first circuit pattern provided on the first substrate **12** and a first interlayer dielectric layer that covers the first circuit pattern. In an embodiment, the first circuit pattern may be a memory circuit, a logic circuit, or a combination thereof, any of which circuits includes one or more transistors. Alternatively, the first circuit pattern may include a passive element, such as a resistor, an inductor, or a capacitor.

(15) Referring to FIGS. **1** and **2**, the first pads **20** may be disposed on the first circuit layer **14**. The first pads **20** may be disposed on a top surface of the first circuit layer **14**. The first pad **20** may have a width that is constant irrespective of a distance from the first substrate **12**. Alternatively, differently from that shown in FIG. **1**, in an embodiment the first pad **20** may have a width that decreases with decreasing distance from the first substrate **12**. The first pad **20** may have a thickness that is substantially uniform. For example, the first pad **20** may have a plate shape. According to some embodiments, the first pad **20** may have a T-shaped cross-section including a via part and an overlying pad part that are integrally connected together. In an embodiment, the first pad **20** may have a circular planar shape. Alternatively, the first pad **20** may have a circular planar shape or a tetragonal planar shape. However, embodiments of the present inventive concept are not necessarily limited thereto, and the planar shape of the first pad **20** may be variously changed. In an embodiment, the first pad **20** may have a width ranging from about 2 μm to about 30 μm . However, embodiments of the present inventive concept are not necessarily limited thereto. The first pads **20** may include a metallic material. For example, in an embodiment the first pads **20** may include copper (Cu).

(16) The first pads **20** may be electrically connected to the first circuit pattern of the first circuit layer **14**. For example, as shown in FIG. **1**, the first circuit layer **14** may include a first connection line **15** disposed therein. The first connection line **15** may be a through via that vertically penetrates the first interlayer dielectric layer in the first circuit layer **14**. The first connection line **15** may vertically extend through the first circuit layer **14** to be coupled to the first pads **20**. The first connection line **15** may electrically connect the first circuit pattern to the first pads **20**. However, embodiments of the present inventive concept are not necessarily limited thereto and various conductive patterns may be provided for connection between the first circuit pattern and the first connection line **15**. For example, differently from that shown in FIG. **1**, in an embodiment the first connection line **15** may be an under-pad pattern or a redistribution pattern disposed in a dielectric pattern in the first circuit layer **14**. In this embodiment, various conductive patterns may be included for connection between the first circuit pattern and the first connection line **15**. However, embodiments of the present inventive concept are not necessarily limited thereto. For example, the first circuit layer **14** may be provided in various forms if necessary, and electrical connections of the first pads **20** and the first circuit layer **14** may be achieved through various configurations if necessary.

(17) The first dielectric layer **16** may be disposed on the first circuit layer **14**. On the top surface of the first circuit layer **14**, the first dielectric layer **16** may surround the first pads **20**. The top surfaces of the first pads **20** may be exposed by the first dielectric layer **16**. For example, when viewed in a plan view, the first dielectric layer **16** may surround, but may not cover the first pads **20**. A top surface of the first dielectric layer **16** may be located at a different level from that of the top surfaces of the first pads **20**. For example, the top surface of the first dielectric layer **16** may be disposed farther from the first substrate **12** than the top surfaces of the first pads **20**. In such a configuration, the top surfaces of the first pads **20** may have shapes that are recessed towards the first substrate **12** from the top surface of the first dielectric layer **16**. Therefore, the first pads **20** may be disposed thereon with first recesses **RS1** that are defined by the first dielectric layer **16** and the first pads **20**. For example, on one of the first pads **20**, the first recess **RS1** may be surrounded by the first dielectric layer **16**. In an embodiment, the first dielectric layer **16** may include oxide, nitride, or oxynitride of a material included in the first substrate **12** or the first circuit layer **14**. In an embodiment, the first dielectric layer **16** may include a dielectric material, such as silicon oxide (SiO), silicon nitride (SiN), silicon oxynitride (SiON), or silicon carbonitride (SiCN). For example, the first dielectric layer **16** may include silicon oxide (SiO).

(18) The first pad **20** may have a damascene structure in the first dielectric layer **16**. For example, the first pads **20** may further include first seed/barrier patterns **22** that cover lateral and bottom surfaces of the first pads **20**. The first seed/barrier patterns **22** may conformally cover the lateral and bottom surfaces of the first pads **20**. The first seed/barrier patterns **22** may be interposed between the first pads **20** and the first dielectric layer **16** and between the first pads **20** and the first circuit layer **14**. In an embodiment, a gap between the first pads **20** and the first dielectric layer **16**, or a thickness of each of the first seed/barrier patterns **22** may be in a range of about 50 Å to about 1,000 Å. In an embodiment in which the first seed/barrier patterns **22** are used as seed patterns, the first seed/barrier patterns **22** may include metal, such as gold (Au). In an embodiment in which the first seed/barrier patterns **22** are used as barrier patterns, the first seed/barrier patterns **22** may include either metal such as titanium (Ti) and tantalum (Ta), or metal nitride such as titanium nitride (TiN) and tantalum nitride (TaN).

(19) Referring back to FIG. 1, the upper structure **30** may be disposed on the lower structure **10**. The upper structure **30** may include a second substrate **32**, a second circuit layer **34**, a second dielectric layer **36**, and second pads **40**.

(20) In an embodiment, the second substrate **32** may be a semiconductor substrate, such as a semiconductor wafer. For example, the second substrate **32** may be a bulk silicon substrate, a silicon on insulator (SOI) substrate, a germanium substrate, a germanium on insulator (GOI) substrate, a silicon-germanium (SiGe) substrate, or an epitaxial film substrate obtained by performing a selective epitaxial growth (SEG) process. The second substrate **32** may include, for example, at least one compound selected from silicon (Si), germanium (Ge), silicon germanium (SiGe), gallium arsenic (GaAs), indium gallium arsenic (InGaAs), aluminum gallium arsenic (AlGaAs), and a mixture thereof. Alternatively, the second substrate **32** may be a dielectric substrate, such as a printed circuit board (PCB).

(21) The second circuit layer **34** may be disposed on the second substrate **32**. As shown in FIG. 1, the second circuit layer **34** may be disposed on a bottom surface of the second substrate **32**. The second circuit layer **34** may include a second circuit pattern disposed on the second substrate **32** and a second interlayer dielectric layer that covers the second circuit pattern. In an embodiment, the second circuit pattern may be a memory circuit, a logic circuit, or a combination thereof, any of which circuits includes one or more transistors. Alternatively, the second circuit pattern may include a passive element, such as a resistor, an inductor, or a capacitor.

(22) Referring back to FIGS. 1 and 2, the second pads **40** may be disposed on the second circuit layer **34**. The second pads **40** may be disposed on a bottom surface of the second circuit layer **34**. The second pad **40** may have a width that is constant irrespective of a distance from the second

substrate **32**. Alternatively, differently from that shown in FIG. **1**, in an embodiment the second pad **40** may have a width that decreases with decreasing distance from the second substrate **32**. The second pad **40** may have a thickness that is substantially uniform. For example, the second pad **40** may have a plate shape. According to some embodiments, the second pad **40** may have a T-shaped cross-section including a via part and an overlying pad part that are integrally connected together. The second pad **40** may have a circular planar shape. Alternatively, the second pad **40** may have a circular planar shape or a tetragonal planar shape. However, embodiments of the present inventive concept are not necessarily limited thereto, and the planar shape of the second pad **40** may be variously changed if necessary. The second pad **40** may have a width ranging from about 2 μm to about 30 μm . However, embodiments of the present inventive concept are not necessarily limited thereto. A material included in the second pads **40** may be the same as that included in the first pads **20**. The second pads **40** may include a metallic material. For example, the second pads **40** may include copper (Cu).

(23) The second pads **40** may be electrically connected to the second circuit pattern of the second circuit layer **34**. For example, as shown in FIG. **1**, the second circuit layer **34** may include a second connection line **35** disposed therein. The second connection line **35** may be a through via that vertically penetrates the second interlayer dielectric layer in the second circuit layer **34**. The second connection line **35** may vertically extend in the second circuit layer **34** to be coupled to the second pads **40**. The second connection line **35** may electrically connect the second circuit pattern to the second pads **40**. However, embodiments of the present inventive concept are not necessarily limited thereto and various conductive patterns may be provided for connection between the second circuit pattern and the second connection line **35**. For example, differently from that shown in FIG. **1**, the second connection line **35** may be an under-pad pattern or a redistribution pattern included in a dielectric pattern in the second circuit layer **34**. In this embodiment, various conductive patterns may be included for connection between the second circuit pattern and the second connection line **35**. However, embodiments of the present inventive concept are not necessarily limited thereto. For example, the second circuit layer **34** may be provided in various forms and electrical connections of the second pads **40** and the second circuit layer **34** may be achieved through various configurations.

(24) The second dielectric layer **36** may be disposed on the second circuit layer **34**. On the bottom surface of the second circuit layer **34**, the second dielectric layer **36** may surround the second pads **40**. The bottom surfaces of the second pads **40** may be exposed by the second dielectric layer **36**. For example, when viewed in a plan view, the second dielectric layer **36** may surround, but may not cover the second pads **40**. The second dielectric layer **36** may have a bottom surface located at a different level from that of the bottom surfaces of the second pads **40**. For example, the bottom surface of the second dielectric layer **36** may be disposed farther from the second substrate **32** than the bottom surfaces of the second pads **40**. In such a configuration, the bottom surfaces of the second pads **40** may have shapes that are recessed towards the second substrate **32** from the bottom surface of the second dielectric layer **36**. Therefore, the second pads **40** may include second recesses RS2 that are defined by the second dielectric layer **36** and the second pads **40**. For example, on one of the second pads **40**, the second recess RS2 may be surrounded by the second dielectric layer **36**. In an embodiment, the second dielectric layer **36** may include oxide, nitride, or oxynitride of a material included in the second substrate **32** or the second circuit layer **34**. For example, the second dielectric layer **36** may include a dielectric material, such as silicon oxide (SiO), silicon nitride (SiN), silicon oxynitride (SiON), or silicon carbonitride (SiCN). For example, the second dielectric layer **36** may include silicon oxide (SiO).

(25) The second pad **40** may have a damascene structure in the second dielectric layer **36**. For example, the second pads **40** may further include second seed/barrier patterns **42** that cover lateral and top surfaces of the second pads **40**. The second seed/barrier patterns **42** may conformity cover the lateral and top surfaces of the second pads **40**. The second seed/barrier patterns **42** may be interposed between the second pads **40** and the second dielectric layer **36** and between the second

pads **40** and the second circuit layer **34**. In an embodiment, a gap between the second pads **40** and the second dielectric layer **36**, or a thickness of each of the second seed/barrier patterns **42** may be in a range of about 50 Å to about 1,000 Å. In an embodiment in which the second seed/barrier patterns **42** are used as seed patterns, the second seed/barrier patterns **42** may include metal, such as gold (Au). In an embodiment in which the second seed/barrier patterns **42** are used as barrier patterns, the second seed/barrier patterns **42** may include either metal such as titanium (Ti) and tantalum (Ta), or metal nitride such as titanium nitride (TiN) and tantalum nitride (TaN).

(26) The upper structure **30** may be disposed on the lower structure **10**. The first pads **20** of the lower structure **10** may be vertically aligned with the second pads **40** of the upper structure **30**. The lower structure **10** and the upper structure **30** may be in direct contact with each other.

(27) The first dielectric layer **16** of the lower structure **10** and the second dielectric layer **36** of the upper structure **30** may be bonded at an interface between the lower structure **10** and the upper structure **30**. In this embodiment, the first dielectric layer **16** and the second dielectric layer **36** may constitute a hybrid bonding of oxide, nitride, or oxynitride. In this description, the term “hybrid bonding” may denote a bonding in which two components of the same kind are merged at an interface therebetween. For example, the first dielectric layer **16** and the second dielectric layer **36** may have a continuous configuration, and an invisible interface IF may be provided between the first dielectric layer **16** and the second dielectric layer **36**. However, embodiments of the present inventive concept are not necessarily limited thereto. For example, the first dielectric layer **16** and the second dielectric layer **36** may be formed of the same material, and thus no interface may be present between the first dielectric layer **16** and the second dielectric layer **36**. Thus, the first dielectric layer **16** and the second dielectric layer **36** may be provided as a single component. For example, the first dielectric layer **16** and the second dielectric layer **36** may be combined to form a single unitary body. However, embodiments of the present inventive concept are not necessarily limited thereto. For example, the first dielectric layer **16** and the second dielectric layer **36** may be formed of different materials. The first dielectric layer **16** and the second dielectric layer **36** may not have a continuous configuration, and the interface IF may be visible between the first dielectric layer **16** and the second dielectric layer **36**. The first dielectric layer **16** and the second dielectric layer **36** may not be coupled to each other, and may each be provided in an individual component. The following description will focus on the embodiment of FIGS. **1** and **2** for convenience of explanation.

(28) The upper structure **30** may be connected to the lower structure **10**. For example, the lower structure **10** and the upper structure **30** may be in direct contact with each other. The first pads **20** of the lower structure **10** may be electrically connected to the second pads **40** of the upper structure **30**.

(29) Interfacial layers **50** may be positioned between the first pads **20** and the second pads **40**. Each of the interfacial layers **50** may be interposed between one of the first pads **20** and one of the second pads **40**. The interfacial layers **50** may separate the first pads **20** from the second pads **40**. The interfacial layers **50** may be in direct contact with the top surfaces of the first pads **20** and with the bottom surfaces of the second pads **40**. For example, in an embodiment entireties of the top surfaces of the first pads **20** may be in direct contact with entireties of bottom surfaces **50a** of the interfacial layers **50**, and entireties of the bottom surfaces of the second pads **40** may be in direct contact with entireties of top surfaces **50b** of the interfacial layers **50**. In this embodiment, entireties of interfaces between the first pads **20** and the interfacial layers **50** may be used in bonding between the first pads **20** and the interfacial layers **50**, and entireties of interfaces between the second pads **40** and the interfacial layers **50** may be used in bonding between the second pads **40** and the interfacial layers **50**.

(30) According to some embodiments of the present inventive concept, the interfacial layers **50** and the first and second pads **20** and **40** may be entirely bonded without impurities therebetween. Therefore, the interfacial layers **50** may be strongly bonded to the first and second pads **20** and **40**,

and a semiconductor device may increase structural stability.

(31) In addition, neither empty spaces nor other materials may be interposed between the interfacial layers **50** and the first and second pads **20** and **40**. Therefore, a small interfacial resistance may be provided between the interfacial layers **50** and the first and second pads **20** and **40**, and a semiconductor device may increase in electrical properties and operating stability.

(32) The overall bonding between the interfacial layers **50** and the first and second pads **20** and **40** may be achieved by a method of fabricating a semiconductor device according to some embodiments of the present inventive concept, and this will be further discussed in detail below.

(33) The first circuit layer **14** and the second circuit layer **34** may be electrically connected through the first pads **20**, the interfacial layers **50**, and the second pads **40**. The bottom surfaces **50a** of the interfacial layers **50** may be closer to the first substrate **12** than the top surface of the first dielectric layer **16**. For example, the bottom surfaces **50a** of the interfacial layers **50** may be located at a lower level than that of the interface IF between the first dielectric layer **16** and the second dielectric layer **36**. In such a configuration, lower portions of the interfacial layers **50** may fill the first recesses RS1. The top surfaces **50b** of the interfacial layers **50** may be closer to the second substrate **32** than the bottom surface of the second dielectric layer **36**. For example, the top surfaces **50b** of the interfacial layers **50** may be located at a higher level than that of the interface IF between the first dielectric layer **16** and the second dielectric layer **36**. In such a configuration, upper portions of the interfacial layers **50** may fill the second recesses RS2. As the upper structure **30** and the lower structure **10** are bonded to each other, the first recesses RS1 and the second recesses RS2 may be connected to form internal spaces, and the interfacial layers **50** may be positioned in the internal spaces.

(34) In an embodiment, the interfacial layer **50** may have a thickness that is substantially uniform. For example, the interfacial layer **50** may have a plate shape. A material included in the interfacial layers **50** may be the same as that included in the first pads **20** and that included in the second pads **40**. The interfacial layers **50** may include a metallic material. For example, the interfacial layers **50** may include copper (Cu).

(35) Referring to FIGS. **1**, **2** and **3**, in an embodiment an average grain size of the first pads **20** may be substantially the same as or similar to that of the second pads **40**. An average grain size of the interfacial layers **50** may be less than that of the first pads **20** and that of the second pads **40**. In this description, an average grain area may be obtained by dividing a certain area by the number of grains included in the certain area, and the diameter of a circle having the average grain area may be defined as an average grain size. Alternatively, an average grain size may be determined by diameter measurement, ASTM grain size number, or the like. The phrase “average grain sizes of certain components are substantially the same as each other” may mean that a difference in average grain sizes of the components is less than about 10%. For convenience of description, FIG. **3** exaggerates grain sizes, and size ratios between grains and components of a semiconductor device may be different from actual ratios. Although FIG. **3** depicts grain sizes of the first pads **20**, the second pads **40**, and the interfacial layers **50**, this illustration shows by way of example a comparison between grain sizes of the interfacial layers **50** and grain sizes of the first and second pads **20** and **40**. However, embodiments of the present inventive concept are not necessarily limited thereto. As the average grain size of the first and second pads **20** and **40** is different from that of the interfacial layers **50**, the first and second pads **20** and **40** and the interfacial layers **50** may be physically distinguishable from each other even in an embodiment in which the first and second pads **20** and **40** and the interfacial layers **50** are formed of the same material. For example, visible interfaces may be provided between the interfacial layers **50** and the first and second pads **20** and **40**.

(36) The interfacial layers **50** may be in direct contact with the first seed/barrier patterns **22** and the second seed/barrier patterns **42**. For example, uppermost surfaces of the first seed/barrier patterns **22** may be in direct contact with the bottom surfaces **50a** of the interfacial layers **50**, and lowermost

surfaces of the second seed/barrier patterns **42** may be in direct contact with the top surfaces **50b** of the interfacial layers **50**. In an embodiment, the interfacial layer **50** may have a width that is the same as an overall width of the first seed/barrier pattern **22** and an overall width of the second seed/barrier pattern **42**. For example, lateral surfaces of the interfacial layers **50** may be aligned with outer lateral surfaces of the first seed/barrier patterns **22** and lateral surfaces of the second seed/barrier patterns **42**.

(37) In the embodiments that follow, a detailed description of technical features repetitive to those discussed above with reference to FIGS. **1**, **2** and **3** will be omitted, and a difference thereof will be discussed in detail for economy of description. The same reference numerals may refer to the same components as those of the semiconductor package according to some embodiments of the present inventive concept.

(38) FIG. **4** illustrates a cross-sectional view of Section A depicted in FIG. **1**, showing a semiconductor device according to some embodiments of the present inventive concept.

(39) Referring to FIGS. **1** and **4**, the interfacial layers **50** may be disposed between the first pads **20** and the second pads **40**. Each of the interfacial layers **50** may be interposed between one of the first pads **20** and one of the second pads **40**. The interfacial layers **50** may separate the first pads **20** from the second pads **40**. The interfacial layers **50** may be in direct contact with the top surfaces of the first pads **20** and with the bottom surfaces of the second pads **40**. For example, in an embodiment entireties of the top surfaces of the first pads **20** may be in direct contact with of the bottom surfaces **50a** of the interfacial layers **50**, and entireties of the bottom surfaces of the second pads **40** may be in direct contact with entireties of the top surfaces **50b** of the interfacial layers **50**.

(40) The interfacial layer **50** may have a non-uniform thickness. For example, a thickness at a lateral edge portion of the interfacial layer **50** may be less than that at a central portion of the interfacial layer **50**. The thickness of the interfacial layer **50** may increase in a direction from the edge portion of the interfacial layer **50** towards the central portion. As the entireties of the top surfaces of the first pads **20** are in direct contact with the entireties of the bottom surfaces **50a** of the interfacial layers **50**, thicknesses at edge portions of the first pads **20** may be greater than those at central portions of the first pads **20**. As the entireties of the bottom surfaces of the second pads **40** are in direct contact with the entireties of the top surfaces **50b** of the interfacial layers **50**, thicknesses at edge portions of the second pads **40** may be greater those at central portions of the second pads **40**.

(41) FIG. **5** illustrates a cross-sectional view of section A depicted in FIG. **1**, showing a semiconductor device according to some embodiments of the present inventive concept.

(42) Referring to FIG. **5**, the interfacial layers **50** may be disposed between the first pads **20** and the second pads **40**. Each of the interfacial layers **50** may be interposed between one of the first pads **20** and one of the second pads **40**. The interfacial layers **50** may separate the first pads **20** from the second pads **40**. The interfacial layers **50** may be in direct contact with the top surfaces of the first pads **20** and with the bottom surfaces of the second pads **40**. For example, in an embodiment entireties of the top surfaces of the first pads **20** may be in direct contact with an entirety of the bottom surfaces **50a** of the interfacial layers **50**, and entireties of the bottom surfaces of the second pads **40** may be in direct contact with entireties of the top surfaces **50b** of the interfacial layers **50**.

(43) In an embodiment, each of the interfacial layers **50** may contain an impurity. The following will describe a single interfacial layer **50**. The impurity may be contained in a portion **52** of the interfacial layer **50** that is positioned at an intermediate location between the top surface **50b** and the bottom surface **50a** of the interfacial layer **50**. For example, a distance between the portion **52** and the top surface **50b** of the interfacial layer **50** may be substantially the same as or similar to that between the portion **52** and the bottom surface **50a** of the interfacial layer **50**. Alternatively, the portion **52** of the interfacial layer **50** may be located at the same level as that of the interface IF between the first dielectric layer **16** and the second dielectric layer **36**. In FIG. **5**, the portion **52** of the interfacial layer **50** is illustrated in the form of a layer so as to explain a position of the portion

52 of the interfacial layer **50**. However, the portion **52** may not be completely separate from upper and lower portions of the interfacial layer **50**. For example, in an embodiment the impurity may be irregularly and partially contained in the interfacial layer **50**. In this sense, the interfacial layer **50** may partially contain the impurity at a location at the same level as that of the interface IF between the first dielectric layer **16** and the second dielectric layer **36**. A concentration of the impurity may decrease in a direction towards the top surface **50b** of the interfacial layer **50** and in a direction towards the bottom surface **50a** of the interfacial layer **50**. In an embodiment, the impurity may include a nitrogen (N) element. The impurity may be a component that remains in the interfacial layer **50** in a procedure for forming a semiconductor device according to a semiconductor fabrication method in accordance with some embodiments of the present inventive concept, and this will be discussed in detail together with a subsequently described method of fabricating a semiconductor device.

(44) FIGS. **6** and **7** illustrate cross-sectional views of section a depicted in FIG. **1**, showing a semiconductor device according to some embodiments of the present inventive concept.

(45) In embodiments shown in FIGS. **1**, **2**, **3**, **4** and **5** the bottom surfaces **50a** of the interfacial layers **50** are located at a lower level than that of the interface IF between the first dielectric layer **16** and the second dielectric layer **36**, and the top surfaces **50b** of the interfacial layers **50** are located at a higher level than that of the interface IF between the first dielectric layer **16** and the second dielectric layer **36**. However, embodiments of the present inventive concept are not necessarily limited thereto.

(46) Referring to FIG. **6**, the interfacial layers **50** may be positioned solely in the upper structure **30**.

(47) The first dielectric layer **16** may have a top surface located at the same level as that of the top surfaces of the first pads **20**. For example, the top surface of the first dielectric layer **16** may be coplanar with those of the first pads **20**. Thus, the first pads **20** may not include the first recesses RS1 discussed with reference to FIGS. **1**, **2**, **3**, **4**, and **5**.

(48) The second dielectric layer **36** may have a bottom surface located at a different level from that of the bottom surfaces of the second pads **40**. For example, the bottom surface of the second dielectric layer **36** may be disposed farther from the second substrate **32** than the bottom surfaces of the second pads **40**. In such a configuration, the shapes of the bottom surfaces of the second pads **40** may be recessed towards the second substrate **32** from the bottom surface of the second dielectric layer **36**. Therefore, the second pads **40** may include second recesses RS2 that are defined by the second dielectric layer **36** and the second pads **40**. For example, on one of the second pads **40**, the second recess RS2 may be surrounded by the second dielectric layer **36**.

(49) The first circuit layer **14** and the second circuit layer **34** may be electrically connected through the first pads **20**, the interfacial layers **50**, and the second pads **40**. The bottom surfaces **50a** of the interfacial layers **50** may be located at the same level as that of the bottom surface of the second dielectric layer **36**. For example, the bottom surfaces **50a** of the interfacial layers **50** may be coplanar with the interface IF between the first dielectric layer **16** and the second dielectric layer **36**. The top surfaces **50b** of the interfacial layers **50** may be closer to the second substrate **32** than the bottom surface of the second dielectric layer **36**. For example, the top surfaces **50b** of the interfacial layers **50** may be located at a higher level than that of the interface IF between the first dielectric layer **16** and the second dielectric layer **36**. In such a configuration, entireties of the interfacial layers **50** may fill the second recesses RS2.

(50) According to some embodiments, referring to FIG. **7**, the interfacial layers **50** may be positioned in the lower structure **10**.

(51) The first dielectric layer **16** may have a top surface located at a different level from that of the top surfaces of the first pads **20**. For example, the top surface of the first dielectric layer **16** may be disposed farther from the first substrate **12** than the top surface of the first pads **20**. In such a configuration, the shapes of the top surfaces of the first pads **20** may be recessed towards the first

substrate **12** from the top surface of the first dielectric layer **16**. Therefore, the first pads **20** may include first recesses **RS1** that are defined by the first dielectric layer **16** and the first pads **20**. For example, on one of the first pads **20**, the first recess **RS1** may be surrounded by the first dielectric layer **16**.

(52) The second dielectric layer **36** may have a bottom surface located at a same level from that of the bottom surfaces of the second pads **40**. For example, the bottom surface of the second dielectric layer **36** may be coplanar with those of the second pads **40**. In this sense, the second pads **40** may not include the second recesses **RS2** discussed with reference to embodiments of FIGS. **1**, **2**, **3**, **4**, and **5**.

(53) The first circuit layer **14** and the second circuit layer **34** may be electrically connected through the first pads **20**, the interfacial layers **50**, and the second pads **40**. The top surfaces **50b** of the interfacial layer **50** may be located at the same level as that of the top surface of the first dielectric layer **16**. For example, the top surfaces **50b** of the interfacial layers **50** may be coplanar with the interface **IF** between the first dielectric layer **16** and the second dielectric layer **36**. The bottom surfaces **50a** of the interfacial layers **50** may be closer to the first substrate **12** than the top surface of the first dielectric layer **16**. For example, the bottom surfaces **50a** of the interfacial layers **50** may be located at a lower level than that of the interface **IF** between the first dielectric layer **16** and the second dielectric layer **36**. In such a configuration, entireties of the interfacial layers **50** may fill the first recesses **RS1**.

(54) FIG. **8** illustrates a cross-sectional view showing a semiconductor device **1** according to some embodiments of the present inventive concept.

(55) Referring to FIG. **8**, a substrate **100** may be provided. In an embodiment, the substrate **100** may be a package substrate such as a printed circuit board (PCB) or an interposer substrate provided in a package. Alternatively, the substrate **100** may be a semiconductor substrate on which semiconductor elements are formed or integrated. The substrate **100** may include a substrate base layer **110** and a substrate wiring layer **120** disposed on the substrate base layer **110**.

(56) The substrate wiring layer **120** may include first substrate pads **122** exposed on a top surface of the substrate base layer **110** and a substrate protection layer **124** that covers the substrate base layer **110** and surrounds the first substrate pads **122**. The first substrate pads **122** may correspond to the first pads **20** discussed with reference to FIGS. **1**, **2**, **3**, **4**, and **5**. For example, top surfaces of the first substrate pads **122** may be located at a lower level than that of a top surface of the substrate protection layer **124**. For example, the first substrate pads **122** may include recesses that are defined by the first substrate pads **122** and the substrate protection layer **124**. Alternatively, identically or similarly to that discussed with reference to FIG. **6**, the top surfaces of the first substrate pads **122** may be coplanar with the top surface of the substrate protection layer **124**.

(57) The substrate base layer **110** may include second substrate pads **130** that are exposed on a bottom surface of the substrate base layer **110**. The substrate **100** may redistribute a chip stack **CS** which will be discussed below. For example, in an embodiment the first substrate pads **122** and the second substrate pads **130** may be electrically connected through a circuit line in the substrate base layer **110**, and a redistribution circuit may be constituted by the circuit line, the first substrate pads **122**, and the second substrate pads **130**. The first substrate pads **122** and the second substrate pads **130** may include a conductive material, such as metal. In an embodiment, the first substrate pads **122** and the second substrate pads **130** may include, for example, copper (Cu). In an embodiment, the substrate protection layer **124** may include a dielectric material, such as oxide, nitride, or oxynitride of a material included in the substrate base layer **110**. For example, the substrate protection layer **124** may include silicon oxide (SiO). However, embodiments of the present inventive concept are not necessarily limited thereto.

(58) The substrate **100** may have substrate connection terminals **140** disposed on a bottom surface thereof. The substrate connection terminals **140** may be disposed on the second substrate pads **130** of the substrate **100**. The substrate connection terminals **140** may include solder balls or solder

bumps. In accordance with type and arrangement of the substrate connection terminals **140**, the semiconductor device **1** may be a ball grid array (BGA), fine ball grid array (FBGA), or land grid array (LGA). However, embodiments of the present inventive concept are not necessarily limited thereto.

(59) A chip stack CS may be disposed on the substrate **100**. The chip stack CS may include one or more semiconductor chips **200** and **200'** that are stacked on the substrate **100**. In an embodiment, each of the semiconductor chips **200** and **200'** may be a memory chip, such as DRAM, SRAM, MRAM, or flash memory. Alternatively, each of the semiconductor chips **200** and **200'** may be a logic chip. FIG. **8** depicts that a single chip stack CS is disposed. However, embodiments of the present inventive concept are not necessarily limited thereto. In an embodiment in which the chip stack CS is provided in plural, the plurality of chip stacks CS may be spaced apart from each other on the substrate **100**.

(60) A single semiconductor chip **200** may be mounted on the substrate **100**. The semiconductor chip **200** may include a semiconductor material, such as silicon (Si). The semiconductor chip **200** may include a chip base layer **210**, a first chip wiring layer **220** disposed on a side of a front surface of the semiconductor chip **200** from the chip base layer **210**, and a second chip wiring layer **230** disposed on a side of a rear surface of the semiconductor chip **200** from the chip base layer **210**. In this description, the language “front surface” may be defined to indicate an active surface of an integrated element in a semiconductor chip or a surface on which are formed a plurality of pads of a semiconductor chip, and the language “rear surface” may be defined to indicate an opposite surface that faces the front surface.

(61) The first chip wiring layer **220** may include first chip pads **222** on the chip base layer **210** and a first chip protection layer **224** by which the first chip pads **222** are surrounded on the chip base layer **210**. The first chip pads **222** may correspond to the second pads **40** discussed with reference to FIGS. **1**, **2**, **3**, **4** and **5**. For example, bottom surfaces of the first chip pads **222** may be located at a higher level than that of a bottom surface of the first chip protection layer **224**. In such a configuration, the first chip pads **222** may include recesses that are defined by the first chip pads **222** and the first chip protection layer **224**. Alternatively, identically or similarly to that discussed with reference to FIG. **7**, in an embodiment the bottom surfaces of the first chip pads **222** may be coplanar with the bottom surface of the first chip protection layer **224**. The first chip pads **222** may be electrically connected to an integrated element or integrated circuits in the semiconductor chip **200**. According to some embodiments, wiring lines for redistribution may be disposed between the first chip pads **222** and an integrated element in the semiconductor chip **200**. The first chip pads **222** may include a conductive material, such as metal. For example, in an embodiment the first chip pads **222** may include copper (Cu). The first chip protection layer **224** may include a dielectric material. For example, in an embodiment the first chip protection layer **224** may include silicon oxide (SiO).

(62) The second chip wiring layer **230** may include second chip pads **232** disposed on the chip base layer **210** and a second chip protection layer **234** by which the second chip pads **232** are surrounded on the chip base layer **210**. The second chip pads **232** may correspond to the first pads **20** discussed with reference to FIGS. **1**, **2**, **3**, **4** and **5**. For example, top surfaces of the second chip pads **232** may be located at a lower level than that of a top surface of the second chip protection layer **234**. In such a configuration, the second chip pads **232** may include recesses that are defined by the second chip pads **232** and the second chip protection layer **234**. Alternatively, identically or similarly to that discussed with reference to FIG. **6**, the top surfaces of the second chip pads **232** may be coplanar with the top surface of the second chip protection layer **234**. The second chip pads **232** may be electrically connected to the first chip wiring layer **220**. According to some embodiments, the second chip pads **232** may be coupled to the first chip wiring layer **220** by means of through electrodes **240** that vertically penetrate the chip base layer **210**. The second chip pads **232** may include a conductive material, such as metal. For example, in an embodiment the second chip pads

232 may include copper (Cu). The second chip protection layer **234** may include a dielectric material. For example, in an embodiment the second chip protection layer **234** may include silicon oxide (SiO).

(63) The semiconductor chip **200** may be mounted on the substrate **100**. As shown in FIG. **8**, the front surface of the semiconductor chip **200** may be directed towards the substrate **100**, and the semiconductor chip **200** may be electrically connected to the substrate **100**. In this embodiment, the front surface of the semiconductor chip **200**, or a bottom surface of the first chip wiring layer **220**, may be in contact with a top surface of the substrate **100**. For example, the first chip protection layer **224** may be in direct contact with the substrate protection layer **124** of the substrate **100**. The first chip pads **222** of the semiconductor chip **200** may be disposed to correspond to the first substrate pads **122** of the substrate **100**. Identically or similarly to that discussed in FIGS. **1**, **2**, **3**, **4**, **5**, **6** and **7**, the first chip pads **222** of the semiconductor chip **200** and the first substrate pads **122** of the substrate **100** may be bonded through interfacial layers **250** between the first chip pads **222** and the first substrate pads **122**. The interfacial layers **250** may be disposed in spaces defined by recesses formed on the first substrate pads **122** and recesses formed on the first chip pads **222**. The interfacial layers **250** may be in direct contact with the top surfaces of the first substrate pads **122** and with the bottom surfaces of the first chip pads **222**. The bottom surfaces of the interfacial layers **250** may be located at a lower level than that of the top surface of the substrate protection layer **124**, and may also have top surfaces located at a higher level than that of the bottom surface of the first chip protection layer **224**. For example, an interface between the semiconductor chip **200** and the substrate **100** may be located at a level between the top and bottom surfaces of the interfacial layers **250**.

(64) In an embodiment, the semiconductor chip **200** may be provided in plural. For example, another semiconductor chip **200** may be mounted on one semiconductor chip **200**. The front surface of the another semiconductor chip **200** may be directed towards the one semiconductor chip **200**. In this embodiment, the front surface of the another semiconductor chip **200** may be in direct contact with the rear surface of the one semiconductor chip **200**. For example, the first chip wiring layer **220** of the another semiconductor chip **200** may be in direct contact with the second chip wiring layer **230** of the one semiconductor chip **200**. For example, the semiconductor chips **200** may be stacked to allow the first chip protection layer **224** and the second chip protection layer **234** to directly contact each other.

(65) The first chip pads **222** of the semiconductor chip **200** may be disposed to correspond to the second chip pads **232** of another semiconductor chip **200** that overlies the first chip pads **222** of the semiconductor chip **200**. Identically or similarly to that discussed with reference to FIGS. **1**, **3**, **4**, **5**, **6** **7**, the first chip pads **222** and the second chip pads **232** of adjacent semiconductor chips **200** may be bonded through the interfacial layers **250** between the first and second chip pads **222** and **232**. The interfacial layers **250** may be disposed in spaces defined by recesses formed on the first chip pads **222** and recesses formed on the second chip pads **232**. The interfacial layers **250** may be in direct contact with the bottom surfaces of the first chip pads **222** and with the top surfaces of the second chip pads **232**. In an embodiment, an interface between the semiconductor chips **200** may be located at a level between the top and bottom surfaces of the interfacial layers **250**. The semiconductor chips **200** may be electrically connected through the first chip pads **222** and the second chip pads **232**. As described above, a plurality of semiconductor chips **200** and **200'** may be stacked on the substrate **100**. While an embodiment shown in FIG. **8** includes four semiconductor chips **200**, **200'**, embodiments of the present inventive concept are not necessarily limited thereto and the number of the semiconductor chips **200**, **200'** may vary.

(66) An uppermost semiconductor chip **200'** of the semiconductor chips **200** and **200'** included in the chip stack CS may have a configuration partially different from those of remaining semiconductor chips **200**. For example, the uppermost semiconductor chip **200'** may have neither the second chip wiring layer **230** nor the through electrodes **240**.

(67) A molding layer **300** may be disposed on the substrate **100**. In an embodiment, the molding layer **300** may cover the top surface of the substrate **100**. The molding layer **300** may surround the chip stack CS. For example, the molding layer **300** may cover lateral surfaces of the semiconductor chips **200** and **200'**. The molding layer **300** may protect the chip stack CS. The molding layer **300** may include a dielectric material. For example, in an embodiment the molding layer **300** may include an epoxy molding compound (EMC). Differently from that shown in FIG. **8**, in an embodiment the molding layer **300** may be formed to cover the chip stack CS. For example, the molding layer **300** may cover a rear surface of the uppermost semiconductor chip **200'**.

(68) It is illustrated in FIG. **8** that the semiconductor chips **200** are mounted on the substrate **100**. However, embodiments of the present inventive concept are not necessarily limited thereto. According to some embodiments, the semiconductor chips **200** may be mounted on a base semiconductor chip. The base semiconductor chip may be a wafer-level semiconductor substrate formed of a silicon semiconductor. The base semiconductor chip may include an integrated circuit. For example, the integrated circuit may include a memory circuit, a logic circuit, or a combination thereof.

(69) FIG. **9** illustrates a plan view showing a semiconductor device according to some embodiments of the present inventive concept. FIG. **10** illustrates a cross-sectional view taken along line I-I' of FIG. **9**, showing a semiconductor device **2** according to some embodiments of the present inventive concept.

(70) Referring to FIGS. **9** and **10**, the semiconductor device **2** may be a memory device. The semiconductor device **2** may have a chip-to-chip (C2C) structure. The chip-to-chip (C2C) structure may be fabricated by forming on a first wafer an upper chip including a cell array structure CAS; forming, on a second wafer different from the first wafer, a lower chip including a peripheral circuit structure PS, and then using a bonding method to connect the upper and lower chips to each other. For example, the bonding method may include electrically connecting a bonding metal formed on an uppermost metal layer of the upper chip to a bonding metal formed on an uppermost metal layer of the lower chip.

(71) Each of the cell array structure CAS and the peripheral circuit structure PS of the semiconductor device **2** may include an external pad bonding area PA, a word-line bonding area WLBA, and a bit-line bonding area BLBA.

(72) The semiconductor device **2** includes a first substrate **12**. In an embodiment, the first substrate **12** may be formed of a semiconductor material, and may be a silicon (Si) substrate, a silicon-germanium (SiGe) substrate, a germanium (Ge) substrate, or a monocrystalline epitaxial layer grown on a monocrystalline silicon substrate. For example, the first substrate **12** may be a silicon substrate. In addition, the first substrate **12** may include a semiconductor doped with impurities having a first conductivity type (e.g., p-type) and/or an intrinsic semiconductor doped with no impurity.

(73) According to some embodiments, the cell array structure CAS may be disposed on the first substrate **12**, and may include stack structures ST, vertical structures VS, and connection structures CPLG, CL, WPLG, and PCL. For example, the first substrate **12** and the cell array structure CAS may correspond to the lower structure **10** discussed with reference to FIG. **1**, and a portion of the cell array structure CAS may correspond to the first circuit layer **14** discussed with reference to FIG. **1**.

(74) On the first substrate **12**, the stack structures ST may extend lengthwise in parallel in a first direction D1 and may be arranged spaced apart from each other in a second direction D2. Each of the stack structures ST may include electrodes EL vertically stacked on the first substrate **12** and dielectric layers ILD interposed between the electrodes EL. The dielectric layers ILD of the stack structures ST may have thicknesses that can be changed depending on properties of a semiconductor memory device. For example, in some embodiments one or more of the dielectric layers ILD may be formed thicker than others of the dielectric layers ILD. In an embodiment, the

dielectric layers ILD may include silicon oxide (SiO). In an embodiment, the electrodes EL may include a conductive material including, for example, a semiconductor layer, a metal silicon layer, a metal layer, a metal nitride layer, or a combination thereof.

(75) The stack structures ST may extend along the first direction D1 from the bit-line bonding area BLBA to the word-line bonding area WLBA, and may have a stepwise structure on the word-line bonding area WLBA. For example, the electrodes EL of the stack structures ST may have lengths in the first direction D1 that decrease with increasing distance from the first substrate 12. The stack structures ST may have variously shaped stepwise structures on the word-line bonding area WLBA.

(76) In some embodiments, the semiconductor device 2 may be a three-dimensional NAND Flash memory device, and cell strings may be integrated on the first substrate 12. In this embodiment, the stack structures ST may be configured such that uppermost and lowermost electrodes EL may be used as gate electrodes of selection transistors. For example, the topmost electrode EL may serve as a gate electrode of a string selection transistor that controls an electrical connection between a bit line BL and the vertical structures VS, and the bottommost electrode EL may serve as a gate electrode of a ground selection transistor that controls an electrical connection between a common source line and the vertical structures VS. Other electrodes EL between the uppermost and lowermost electrodes EL may serve as control gate electrodes of memory cells and as word lines that connect the control gate electrodes to each other.

(77) On the bit-line bonding area BLBA, the vertical structures VS may penetrate the stack structures ST to come into direct contact with the first substrate 12. The vertical structures VS may be electrically connected to the first substrate 12. In an embodiment, when viewed in a plan view, the vertical structures VS may be arranged in a straight or zigzag fashion along one direction. In addition, in an embodiment on the word-line bonding area WLBA or the external pad bonding area PA, dummy vertical structures may be disposed which have substantially the same structure as that of the vertical structures VS.

(78) In an embodiment, the vertical structures VS may include a semiconductor material, such as silicon (Si), germanium (Ge), or a mixture thereof. Additionally or alternatively, the vertical structures VS may include an impurity-doped semiconductor or an impurity-undoped intrinsic semiconductor. The vertical structures VS including a semiconductor material may be used as channels of selection transistors and channels of memory cell transistors. The vertical structures VS may have their bottom surfaces between top and bottom surfaces of the first substrate 12. The vertical structures VS may each have, on its upper end, a contact pad coupled to a bit-line contact plug BPLG.

(79) Each of the vertical structures VS may include a vertical dielectric pattern VP and a semiconductor pattern SP in direct contact with the first substrate 12. In an embodiment, the semiconductor pattern SP may have a hollow pipe shape or a macaroni shape. The semiconductor pattern SP may have a closed shape at a bottom end thereof, and a buried dielectric pattern VI may fill an inside of the semiconductor pattern SP. The semiconductor pattern SP may be in direct contact with the top surface of the first substrate 12. In an embodiment, the semiconductor pattern SP may be undoped or doped with impurities having a conductivity type is the same as that of the first substrate 12. In an embodiment, the semiconductor pattern SP may be polycrystalline or monocrystalline.

(80) The vertical dielectric pattern VP may be disposed between the stack structure ST and the vertical structure VS. The vertical dielectric pattern VP may extend in a third direction D3 and may surround a sidewall of the vertical structure VS. For example, in an embodiment the vertical dielectric pattern VP may have a macaroni shape or a pipe shape having top and bottom ends that are open. The vertical dielectric pattern VP may be formed of a single thin layer or a plurality of thin layers. In some embodiments of the present inventive concept, the vertical dielectric pattern VP may be a portion of a data storage layer. For example, the vertical dielectric pattern VP may

include a tunnel dielectric layer, a charge storage layer, and a blocking dielectric layer. These layers may constitute a data storage layer of a NAND Flash memory device. For example, the charge storage layer may be a trap dielectric layer, a floating gate electrode, or a dielectric layer including conductive nano-dots. In an embodiment, the charge storage layer may include at least one compound selected from silicon nitride (SiN) silicon oxynitride (SiON), silicon-rich nitride, nano-crystalline silicon, and a laminated trap layer. The tunnel dielectric layer may be one of materials having a band gap that is greater than that of the charge storage layer, and the blocking dielectric layer may be a high-k dielectric layer such as aluminum oxide (Al.sub.2O.sub.3) and hafnium oxide (Hf.sub.2O). Alternatively, the vertical dielectric pattern VP may include a thin layer either for a phase change memory device or for a variable resistance memory device.

(81) The first substrate **12** may have common source regions CSR disposed therein between neighboring stack structures ST. The common source regions CSR may extend lengthwise in the first direction D1 parallel to the stack structures ST. In an embodiment, the common source regions CSR may be formed by doping the first substrate **12** with impurities having a second conductivity type. In an embodiment, the common source regions CSR may include, for example, n-type impurities (e.g., arsenic (As) or phosphorous (P)).

(82) A common source plug CSP may be coupled to the common source region CSR. A sidewall dielectric spacer SSP may be interposed between the common source plug CSP and the stack structures ST. In a read or program mode of a three-dimensional NAND Flash memory device, a ground voltage may be applied through the common source plug CSP to the common source region CSR.

(83) A first buried dielectric layer **450** may be disposed on the first substrate **12** and may cover stepwise ends of the electrodes EL. A first interlayer dielectric layer **451** may cover top surfaces of the vertical structures VS, and the first interlayer dielectric layer **451** may be disposed thereon with a second interlayer dielectric layer **453** that covers a top surface of the common source plug CSP.

(84) Bit lines BL that extend lengthwise in the second direction D2 and run across the stack structures ST may be disposed on the second interlayer dielectric layer **453**. Each of the bit lines BL may be electrically connected through the bit-line contact plug BPLG to the vertical structure VS. The bit lines BL may correspond to pads for electrical connection with the peripheral circuit structure PS which will be discussed. The bit lines BL may have bit-line pads BLP. The bit-line pads BLP may be the same as or similar to the first pads **20** discussed with reference to FIGS. **1**, **2**, **3**, **4** **5**. For example, top surfaces of the bit-line pads BLP may be located at a lower level than that of a top surface of a third interlayer dielectric layer **455** which will be discussed below. In such a configuration, bit-line pads BLP may include recesses that are defined by the bit-line pads BLP and the third interlayer dielectric layer **455**.

(85) A connection line structure that electrically connects the cell array structure CAS to the peripheral logic structure PS may be disposed on the stepwise shaped ends of the stack structures ST. The connection line structure may include cell contact lugs CPLG that penetrate the first buried dielectric layer **450** and the first and second interlayer dielectric layers **451** and **453** to come into connection with (e.g., directly contact) corresponding ends of the electrodes EL, and may also include connection lines CL that are disposed on the second interlayer dielectric layer **453** to come into connection with (e.g., directly contact) corresponding cell contact plugs CPLG. In addition, the connection line structure may include well contact plugs WPLG coupled to well pickup regions PUR in the external pad bonding area PA of the first substrate **12**, and may also include peripheral connection lines PCL connected to the well contact plugs WPLG. The bit lines BL, the connection lines CL, and the peripheral connection lines PCL may constitute a cell array wiring layer **460**.

(86) The first substrate **12** may have therein the well pickup regions PUR that are disposed adjacent to opposite ends of each of the stack structures ST. In an embodiment, the well pickup regions PUR may have the same conductivity type as that of the first substrate **12** and, an impurity concentration of the well pickup regions PUR may be greater than that of the first substrate **12**. For example, in

an embodiment the well pickup regions PUR may include heavily doped p-type impurities (e.g., boron (B)). According to some embodiments, in an erase operation of a three-dimensional NAND Flash memory device, an erase voltage may be applied to the well pickup regions PUR through the connection contact plugs PPLG and the well contact plugs WPLG.

(87) A third interlayer dielectric layer **455** that surrounds the bit lines BL, the connection lines CL, and the peripheral connection lines PCL may be disposed on the second interlayer dielectric layer **453**. The third interlayer dielectric layer **455** may expose top surfaces of the bit-line pads BLP, top surfaces of the connection lines CL, and top surfaces of the peripheral connection lines PCL. The third interlayer dielectric layer **455** may be the same as or similar to the first dielectric layer **16** discussed with reference to FIGS. **1**, **2**, **3**, **4**, **5**, **6** and **7**. The bit lines BL, the connection lines CL, and the peripheral connection lines PCL may constitute the cell array wiring layer **460**. The bit lines BL, the connection lines CL, and the peripheral connection lines PCL may correspond to pads of the cell array structure CAS electrically connected to the peripheral circuit structure PS which will be discussed below.

(88) As described above, the cell array structure CAS may be disposed on the first substrate **12**. The peripheral circuit structure PS may be disposed on the cell array structure CAS.

(89) The semiconductor device **2** may include a second substrate **32**. In an embodiment, the second substrate **32** may be a silicon substrate, a silicon-germanium substrate, a germanium substrate, or a monocrystalline epitaxial layer grown on a monocrystalline silicon substrate. For example, the second substrate **32** may be a silicon substrate having a first conductivity type (e.g., p-type), and may include well regions.

(90) The peripheral logic structure PS may include peripheral circuits that are integrated on a front surface of the second substrate **32** and a second buried dielectric layer **550** that covers the peripheral circuits. For example, the second substrate **32** and the peripheral circuit structure PS may correspond to the upper structure **30** discussed with reference to FIG. **1**, and a portion of the peripheral circuit structure PS may correspond to the second circuit layer **34** discussed with reference to FIG. **1**.

(91) In an embodiment, the peripheral circuits may be row and column decoders, a page buffer, and a control circuit, and may include NMOS and PMOS transistors, low-voltage and high-voltage transistors, and a resistor that are integrated on one surface of the second substrate **32**. For example, the peripheral circuits may include a free charge control circuit that controls a plurality of data program steps for a plurality of memory cells and that controls one or more of a plurality of cell strings. For example, a device isolation layer **511** that defines active regions may be disposed on the second substrate **32**. Peripheral gate electrodes **523** may be disposed on the active region of the second substrate **32**. Gate dielectric layers may be interposed between the peripheral gate electrodes **523** and the second substrate **32**. The second substrate **32** may include source/drain regions **521** positioned on opposite sides of the peripheral gate electrode **523**.

(92) A peripheral circuit wiring layer **530** may be connected to the peripheral circuits on the second substrate **32**. The peripheral circuit wiring layer **530** may include peripheral circuit wiring lines **533** and peripheral circuit contact plugs **531**. The peripheral circuit wiring lines **533** may be electrically connected through the peripheral circuit contact plugs **531** to the peripheral circuits. For example, the peripheral circuit contact plugs **531** and the peripheral circuit wiring lines **533** may be coupled to the NMOS and PMOS transistors.

(93) The second buried dielectric layer **550** may cover the peripheral gate electrodes **523**, the peripheral circuit contact plugs **531**, and the peripheral circuit wiring lines **533**. The peripheral circuit wiring layer **530** may further include exposure wiring lines **535** that are exposed on a bottom surface of the second buried dielectric layer **550**. The exposure wiring lines **535** may correspond to pads for electrically connecting the peripheral circuit structures PS to the cell array structure CAS. The exposure wiring lines **535** may have peripheral circuit pads PCP. The peripheral circuit pads PCP may be the same as or similar to the second pads **40** discussed with reference to

FIGS. 1, 2, 3, 4 and 5. For example, the peripheral circuit pads PCP may have widths less than those of the bit-line pads BLP, and thicknesses greater than those of the bit-line pad BLP. For example, the peripheral circuit pads PCP may have bottom surfaces at a higher level than that of the bottom surface of the second buried dielectric layer 550. The peripheral circuit pads PCP may include recesses that are defined by the peripheral circuit pads PCP and the second buried dielectric layer 550. The second buried dielectric layer 550 may include a plurality of stacked dielectric layers. For example, the second buried dielectric layer 550 may include one or more compounds selected from silicon oxide (SiO), silicon nitride (SiN), silicon oxynitride (SiON), and a low-k dielectric material. For example, in an embodiment the peripheral circuit wiring lines 533 and the peripheral circuit contact plugs 531 may be formed of tungsten having a resistance that is relatively higher, and the exposure wiring lines 535 may be formed of copper whose resistance is relatively lower.

(94) The peripheral circuit wiring lines 533 of a single layer are illustrated and explained in this description. However, embodiments of the present inventive concept are not necessarily limited thereto. For example, in an embodiment the peripheral circuit wiring lines 533 may be provided in a plurality of layers, and the peripheral circuit wiring lines 533 in one layer may be stacked on the peripheral circuit wiring lines 533 in another layer. In this embodiment, one or more of the peripheral circuit wiring lines 533 may be formed of aluminum having a resistance that is less than that of copper of which the exposure wiring lines 535 are formed.

(95) The cell array structure CAS and the peripheral circuit structure PS may be in direct contact with each other. For example, as shown in FIG. 10, the cell array wiring layer 460 of the cell array structure CAS may be in direct contact with the peripheral circuit wiring layer 530 of the peripheral circuit structure PS. For example, the third interlayer dielectric layer 455 may be in direct contact with the second buried dielectric layer 550, and the exposure wiring lines 535 may be connected to one or more of the peripheral connection lines PCL, the connection lines CL, and the bit lines BL. The bit-line pads BLP and the peripheral circuit pads PCP may be in contact through interfacial layers 600 disposed between the bit-line pads BLP and the peripheral circuit pads PCP. The interfacial layers 600 may be provided in spaces defined by recesses formed on the bit-line pads BLP and recesses formed on the peripheral circuit pads PCP. The interfacial layers 600 may be in direct contact with the top surfaces of the bit-line pads BLP and with the bottom surfaces of the peripheral circuit pads PCP. An interface between the cell array structure CAS and the peripheral circuit structure PS may be located at a level between top and bottom surfaces of the interfacial layers 600. The cell array structure CAS and the peripheral circuit structure PS may be electrically connected through the bit-line pads BLP and the peripheral circuit pads PCP.

(96) FIGS. 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24 and 25 illustrate cross-sectional views showing a method of fabricating a semiconductor device according to some embodiments of the present inventive concept. FIGS. 11, 12, 13, 14, 15, 16, 17, 19, 21, 23, and 25 are cross-sectional views. FIGS. 18, 20, 22, and 24 are cross-sectional views of section C depicted in FIGS. 17, 19, 21, and 23.

(97) Referring to FIG. 11, a first substrate 12 may be provided. The first substrate 12 may be a semiconductor substrate. A first circuit layer 14 may be formed on the first substrate 12. The first circuit layer 14 may have a first connection line 15 for connecting the first substrate 12 to first pads (see 20 of FIG. 18). A dielectric material may be deposited to form a first dielectric layer 16 on the first circuit layer 14.

(98) Referring to FIG. 12, the first dielectric layer 16 may be patterned to form holes in which first pads (see 20 of FIG. 13) are provided. A first seed/barrier layer 24 may be formed on the first dielectric layer 16. The first seed/barrier layer 24 may be formed to conformally cover the first dielectric layer 16. For example, the first seed/barrier layer 24 may cover a top surface of the first dielectric layer 16 and may also cover bottom and inner lateral surfaces of the holes.

(99) A first conductive layer 26 may be formed on the first dielectric layer 16 and may fill the

holes. In an embodiment, the formation of the first conductive layer **26** may include a plating process in which the first seed/barrier layer **24** is used as a seed. In an embodiment, the first conductive layer **26** may include a metallic material, such as copper (Cu). The first conductive layer **26** may cover the top surface of the first dielectric layer **16**.

(100) Referring to FIG. **13**, a first planarization process may be performed on the first conductive layer **26**. In an embodiment, the first planarization process may include a chemical mechanical polishing (CMP) process. The first planarization process may form first pads **20** and first seed/barrier patterns **22** in the holes. For example, the first conductive layer **26** and the first seed/barrier layer **24** may be partially removed from the top surface of the first dielectric layer **16**, and the top surface of the first dielectric layer **16** may be exposed. The first conductive layer **26** may be over-etched during the first planarization process. For example, after the first planarization process, top surfaces of the first pads **20** may be located at a lower level than that of the top surface of the first dielectric layer **16**. Therefore, the first pads **20** may be provided thereon with first recesses **RS1** that are defined by the first pads **20** and the first dielectric layer **16**. The first recesses **RS1** may have bottom surfaces having a concave shape in a direction towards the first substrate **12**. However, embodiments of the present inventive concept are not necessarily limited thereto, and the bottom surfaces of the first recesses **RS1** may be parallel to a top surface of the first substrate **12**.

(101) According to some embodiments, as shown in FIG. **14**, after the first planarization process, uppermost ends of the first pads **20** may be located at the same level as that of the top surface of the first dielectric layer **16**. The top surfaces of the first pads **20** may have concave shapes in a direction towards the first substrate **12**. However, embodiments of the present inventive concept are not necessarily limited thereto, and an entirety of the top surfaces of the first pads **20** may be coplanar with that of the first dielectric layer **16** in some embodiments.

(102) Referring to FIG. **15**, first surface layers **54** may be formed on the first pads **20**. The first surface layers **54** may fill upper portions of the first recesses **RS1**. The first surface layers **54** may be formed to conformally cover the top surfaces of the first pads **20**, and in accordance with shapes of the bottom surfaces of the first recesses **RS1** and thicknesses of the first surface layers **54**, portions of the upper portions of the first recesses **RS1** may not be filled. In an embodiment, the first surface layers **54** may be formed of the same material as that of the first pads **20**. For example, the first surface layers **54** may include a metallic material, such as copper (Cu).

(103) In an embodiment, a sputtering process may be used to form the first surface layers **54**. The sputtering process may deposit the metallic material on the first pads **20**. In this step, a portion of the metallic material may also be deposited on the top surface of the first dielectric layer **16**. In an embodiment in which the metallic material used in the sputtering process is the same as a material included in the first pads **20**, the metallic material may be deposited as thin films on the first pads **20**. In an embodiment in which the metallic material used in the sputtering process is different from a material included in the first dielectric layer **16**, the metallic material may be difficult to deposit on the first dielectric layer **16**. Therefore, the first dielectric layer **16** may be provided thereon with islands **56** that are formed of the metallic material. Alternatively, the metallic material may not be deposited on the first dielectric layer **16**.

(104) In an embodiment in which the first surface layers **54** and the first pads **20** are formed by using different processes, the first surface layers **54** and the first pads **20** may have different crystalline properties even when the first surface layers **54** and the first pads **20** are formed of the same material. For example, an average grain size of the first surface layers **54** formed by a sputtering process may be less than that of the first pads **20** formed by a plating process.

(105) According to some embodiments, as shown in FIG. **16**, the first surface layers **54** may be formed on the first pads **20** in a resultant structure of FIG. **14**. The first surface layers **54** may fill entire upper portions of the first recesses **RS1**.

(106) Referring to FIGS. **17** and **18**, a second planarization process may be performed on the first dielectric layer **16** and the first surface layers **54**. In an embodiment, the second planarization

process may include a buffing process. The second planarization process may remove the islands **56** on the first dielectric layer **16**. In addition, the first surface layers **54** located at a higher level than that of the top surface of the first dielectric layer **16** may also be removed. As the buffing process having a low grinding force is performed as the second planarization process, neither the first surface layers **54** nor the first pads **20** may be over-etched. For example, neither the first pads **20** nor the first surface layers **54** positioned in the first recesses **RS1** may be partially etched.

(107) In the second planarization process, an oxide layer **58** may be formed on the first surface layers **54**. For example, upper portions of the first surface layers **54** may be oxidized due to an abrasive used in the second planarization process or heat generated in the second planarization process. In an embodiment, the oxide layer **58** may include oxide of a metallic material included in the first surface layers **54**. For example, the oxide layer **58** may include copper oxide (CuO).

(108) Referring to FIGS. **19** and **20**, a first surface treatment process may be performed on the first dielectric layer **16** and the first surface layers **54**. In an embodiment, the first surface treatment process may include argon (Ar) plasma treatment process. The first surface treatment process may remove the oxide layer **58**. Accordingly, top surfaces of the first surface layers **54** may be exposed.

(109) Referring to FIGS. **21** and **22**, a second surface treatment process may be performed on the first dielectric layer **16** and the first surface layers **54**. In an embodiment, the second surface treatment process may include nitrogen (N) plasma treatment process. The second surface treatment process may be performed separately from the first surface treatment process. For example, the second surface treatment process may be executed after the first surface treatment process is completed. After the nitrogen plasma treatment process is performed, a first metal nitride layer **62** may be formed on the top surfaces of the first surface layers **54**, and a first hydroxide layer **64** may be formed on the top surface of the first dielectric layer **16**. For example, nitrogen plasma provided in the nitrogen plasma treatment may react with the top surfaces of the first surface layers **54**, with the result that the first metal nitride layer **62** may be formed. The nitrogen plasma provided in the nitrogen plasma treatment process may react with the top surface of the first dielectric layer **16**, with the result that the first hydroxide layer **64** may be formed. In an embodiment, the first metal nitride layer **62** may include copper nitride (Cu.sub.4N). An average grain size of the first surface layers **54** formed by a sputtering process may be less than that of the first pads **20** formed by a plating process. Copper nitride (Cu.sub.4N) may be more easily formed on the first surface layers **54** than on the first pads **20**. In addition, copper nitride (Cu.sub.4N) may be more easily formed than copper nitride (Cu.sub.3N) on the first surface layers **54**. Since nitrogen nitride (Cu.sub.4N) is more chemically unstable than copper nitride (Cu.sub.3N), the first metal nitride layer **62** formed of copper nitride (Cu.sub.4N) may be easily removed in a subsequent process. The first metal nitride layer **62** may serve as an oxidation barrier layer for preventing oxidation of the first surface layers **54**. The first hydroxide layer **64** may include silicon hydroxide (Si—OH).

(110) A lower structure **10** may be fabricated as described above. An upper structure **30** may be fabricated identically or similarly to the lower structure **10**.

(111) Referring to FIGS. **23** and **24**, the upper structure **30** may be provided on the lower structure **10**. For example, the upper structure **30** may be disposed on the lower structure **10** to allow the first pads **20** to vertically align with the second pads **40**.

(112) Afterwards, the lower structure **10** and the upper structure **30** may be in direct contact with each other. The top surface of the first dielectric layer **16** included in the lower structure **10** may be in direct contact with a bottom surface of a second dielectric layer **36** included in the upper structure **30**. For example, the first hydroxide layer **64** on the first dielectric layer **16** may be in direct contact with a second hydroxide layer **68** on the second dielectric layer **36**. The top surfaces of the first surface layers **54** included in the lower structure **10** may face bottom surfaces of second surface layers **55** included in the upper structure **30**. In this step, based on shapes of the first recesses **RS1** formed on the first pads **20** and shapes of second recesses **RS2** formed on second

pads **40**, air gaps AG may be formed between the first surface layers **54** and the second surface layers **55**. For example, the air gaps AG may be defined by the first metal nitride layers **62** on the first surface layers **54** and second metal nitride layers **66** on the second surface layers **55**.

(113) Referring to FIGS. **1** and **25**, an annealing process may be performed on the lower structure **10** and the upper structure **30**.

(114) In the annealing process, the first surface layers **54** of the lower structure **10** and the second surface layers **55** of the upper structure **30** may be bonded to form interfacial layers **50**. For example, the annealing process may expand the first pads **20** and the second pads **40**. The top surfaces of the first pads **20** may move upwards, and bottom surfaces of the second pads **40** may move downwards. Therefore, the first surface layers **54** and the second surface layers **55** may be in direct contact with each other across the first metal nitride layer **62** and the second metal nitride layer **66**, and the first recesses RS1 and the second recesses RS2 may be removed. The annealing process may remove nitrogen (N) elements from first metal nitride layer **62** and the second metal nitride layer **66**, and the first metal nitride layer **62** and the second metal nitride layer **66** may become portions of the first surface layer **54** and the second surface layer **55**, respectively. The first surface layer **54** and the second surface layer **55** may be combined into a single unitary body. The bonding between first surface layer **54** and the second surface layer **55** may be automatically performed. For example, the first surface layers **54** and the second surface layers **55** may be formed of the same material (e.g., copper (Cu)), and may be combined with each other by an intermetallic hybrid bonding process due to a surface activation at interfaces between the first surface layers **54** and the second surface layers **55** that are in direct contact with each other.

(115) According to some embodiments of the present inventive concept, the bonding process of the lower structure **10** and the upper structure **30** may remove the first recesses RS1 that are formed by the first planarization process performed on the first pads **20**. In addition, easily removable copper nitride (Cu.sub.4N) may be used to form the first and second metal nitride layers **62** and **66** that serve as oxidation barrier layers. Therefore, neither empty spaces nor other material layers may remain between the first pads **20**, the interfacial layers **50**, and the second pads **40**, and the first pads **20** and the second pads **40** may be easily connected to each other. As a result, a failure may be less likely to occur in fabricating a semiconductor device.

(116) According to some embodiments, portions of nitrogen (N) elements that are removed from the first metal nitride layers **62** and the second metal nitride layers **66** may remain in the interfacial layers **50**.

(117) The annealing process may bond the first dielectric layer **16** of the lower structure **10** to the second dielectric layer **36** of the upper structure **30**. For example, in the annealing process, a silicon oxide (SiO) layer may be formed from the first hydroxide layer **64** and the second hydroxide layer **68** that are in direct contact with each other. For example, during the annealing process, the first dielectric layer **16**, the first hydroxide layer **64**, the second hydroxide layer **68**, and the second dielectric layer **36** may all include silicon oxide (SiO), and may be combined into a single unitary body.

(118) FIG. **26** illustrates a cross-sectional view showing a method of fabricating a semiconductor device according to some embodiments of the present inventive concept.

(119) Referring to FIGS. **11**, **12**, **13**, **14**, **15**, **16**, **17**, **18**, **19**, **21** and **22**, the lower structure **10** may be fabricated. The fabrication of the upper structure **30** may be slightly different from the fabrication of the lower structure **10**. For example, the upper structure **30** may be fabricated through the processes of FIGS. **11**, **12**, **13** and **14**. For example, the second surface layers **55** may not be formed on the second pads **40** of the upper structure **30**.

(120) Referring back to FIG. **26**, the upper structure **30** may be provided on the lower structure **10**. For example, the upper structure **30** may be disposed on the lower structure **10** to allow the first pads **20** to vertically align with the second pads **40**.

(121) The lower structure **10** and the upper structure **30** may be in direct contact with each other.

The top surface of the first dielectric layer **16** included in the lower structure **10** may be in direct contact with the bottom surface of the second dielectric layer **36** included in the upper structure **30**. The top surfaces of the first surface layers **54** included in the lower structure **10** may face the bottom surfaces of the second pads **40** included in the upper structure **30**. In this step, based on shapes of the first recesses **RS1** formed on the first pads **20**, the air gaps **AG** may be formed between the first surface layers **54** and the second surface layers **55**. For example, the air gaps **AG** may be defined by the first metal nitride layers **62** on the first surface layers **54** and by the second pads **40**.

(122) An annealing process may be performed on the lower structure **10** and the upper structure **30**.

(123) The annealing process may cause the first surface layers **54** of the lower structure **10** to form the interfacial layers **50**. For example, the annealing process may expand the first pads **20** and the second pads **40**. The top surfaces of the first pads **20** may move upwards, and bottom surfaces of the second pads **40** may move downwards. Therefore, the first surface layers **54** may be in direct contact with the second pads **40**, and the first recesses **RS1** may be removed. In the annealing process, nitrogen (N) elements may be removed from the first metal nitride layer **62**, and the first metal nitride layer **62** may become portions of the first surface layers **54**.

(124) According to some embodiments of the present inventive concept, portions of nitrogen (N) element removed from the first metal nitride layer **62** may remain in the interfacial layers **50**.

(125) The annealing process may bond the first dielectric layer **16** of the lower structure **10** to the second dielectric layer **36** of the upper structure **30**. The first dielectric layer **16** and the second dielectric layer **36** may be combined into a single unitary body.

(126) A semiconductor device according to some embodiments of the present inventive concept may include interfacial layers and pads that are entirely bonded to each other without impurities therebetween. Therefore, the interfacial layers and the pads may be strongly bonded to each other, and the semiconductor device may increase in structural stability. In addition, neither empty spaces nor other material layers may be interposed between the interfacial layers and the pads.

Accordingly, small interfacial resistance may be provided between the interfacial layers and the pads, and the semiconductor device may exhibit an increase in electrical properties and operating stability.

(127) In a method of fabricating a semiconductor device according to some embodiments of the present inventive concept, recesses formed by a first planarization process performed on the pads may be removed in a bonding process of a lower structure and an upper structure. In addition, easily removable copper nitride (Cu.sub.4N) may be used to form metal nitride layers that serve as oxidation barrier layers. Therefore, neither empty spaces nor other material layers may remain between the interfacial layers and the pads, and the pads may be easily connected to each other. Accordingly, a failure may be less likely to occur in fabricating the semiconductor device.

(128) Although the present inventive concept have been described in connection with the some embodiments illustrated in the accompanying drawings, it will be understood by one of ordinary skill in the art that variations in form and detail may be made therein without departing from the spirit and essential feature of the present inventive concept. The above disclosed embodiments should thus be considered illustrative and not restrictive.

Claims

1. A semiconductor device, comprising: a lower structure; and an upper structure on the lower structure, wherein the lower structure includes: a first semiconductor substrate; a first pad on the first semiconductor substrate; and a first dielectric layer on the first semiconductor substrate, the first dielectric layer surrounds the first pad and exposes a top surface of the first pad, wherein the upper structure includes: a second semiconductor substrate; a second pad on the second semiconductor substrate; and a second dielectric layer on the second semiconductor substrate, the

second dielectric layer surrounds the second pad and exposes a bottom surface of the second pad, wherein the first pad and the second pad are bonded to each other across an interfacial layer to couple the upper structure and the lower structure to each other, wherein the first pad, the second pad, and the interfacial layer include a same metallic material, and wherein the first and second pads have a substantially same average grain size as each other and the interfacial layer has a different average grain size than the first and second pads.

2. The semiconductor device of claim 1, wherein: a top surface of the interfacial layer is positioned at a higher level than a level of an interface between the lower structure and the upper structure; and a bottom surface of the interfacial layer is positioned at a lower level than the level of the interface between the lower structure and the upper structure.

3. The semiconductor device of claim 1, wherein: the interfacial layer has a plate shape; and the first pad and the second pad are spaced apart from each other by the interfacial layer.

4. The semiconductor device of claim 3, wherein: a thickness of the first pad is substantially uniform; and a thickness of the second pad is substantially uniform.

5. The semiconductor device of claim 3, wherein a thickness of the interfacial layer is substantially uniform.

6. The semiconductor device of claim 1, wherein the interfacial layer contains a nitrogen (N) element disposed in the metallic material.

7. The semiconductor device of claim 6, wherein a concentration of the nitrogen element in the interfacial layer decreases in directions from the interfacial layer towards a top surface of the interfacial layer and a bottom surface of the interfacial layer.

8. The semiconductor device of claim 1, wherein the metallic material includes copper (Cu).

9. The semiconductor device of claim 1, wherein: the first dielectric layer and the second dielectric layer include oxide; and the first dielectric layer and the second dielectric layer constitute a single unitary body formed of a same material.

10. The semiconductor device of claim 1, wherein: a top surface of the interfacial layer is positioned at a same level as a level of an interface between the lower structure and the upper structure; or a bottom surface of the interfacial layer is positioned at a same level as the level of the interface between the lower structure and the upper structure.

11. The semiconductor device of claim 1, wherein a thickness of an edge portion of the interfacial layer is less than a thickness of a central portion of the interfacial layer.

12. A semiconductor device, comprising: a lower structure; and an upper structure on the lower structure, wherein the lower structure includes: a first semiconductor substrate; a first pad on the first semiconductor substrate; and a first dielectric layer on the first semiconductor substrate, the first dielectric layer surrounds the first pad and exposes a top surface of the first pad, wherein the upper structure includes: a second semiconductor substrate; a second pad on the second semiconductor substrate; and a second dielectric layer on the second semiconductor substrate, the second dielectric layer surrounds the second pad and exposes a bottom surface of the second pad, wherein the first pad and the second pad are bonded to each other across an interfacial layer to couple the upper structure and the lower structure to each other, and wherein an interface between the first dielectric layer and the second dielectric layer is positioned at a level between a top surface and a bottom surface of the interfacial layer.

13. The semiconductor device of claim 12, wherein: the first pad, the second pad, and the interfacial layer include a same metallic material, and the first and second pads have a substantially same average grain size as each other and the interfacial layer has a different average grain size than the first and second pads.

14. The semiconductor device of claim 12, wherein: the interfacial layer has a plate shape; and the first pad and the second pad are spaced apart from each other by the interfacial layer.

15. The semiconductor device of claim 14, wherein: a thickness of the first pad is substantially uniform; and a thickness of the second pad is substantially uniform.

16. The semiconductor device of claim 14, wherein a thickness of the interfacial layer is substantially uniform.

17. The semiconductor device of claim 13, wherein the interfacial layer contains a nitrogen (N) element disposed in the metallic material.

18. The semiconductor device of claim 13, wherein the metallic material includes copper (Cu).

19. The semiconductor device of claim 12, wherein: the first dielectric layer and the second dielectric layer include oxide; and the first dielectric layer and the second dielectric layer constitute a single unitary body formed of a same material.

20. A method of fabricating a semiconductor device, the method comprising forming an upper structure and a lower structure, wherein the forming of the upper and lower structures includes: forming a dielectric layer on a semiconductor substrate; forming a hole in the dielectric layer; performing a plating process, the plating process forming a conductive layer that fills the hole and covers the dielectric layer; performing a planarization process on the conductive layer, the planarization process forming a pad in the hole, the pad includes a recess positioned on a top surface of the pad and extending towards the semiconductor substrate; performing a deposition process, the deposition process forming a surface layer on the top surface of the pad; performing an argon (Ar) plasma treatment process on the surface layer and the dielectric layer; and after performing the argon plasma treatment process, performing a nitrogen (N) plasma treatment process on the surface layer and the dielectric layer; placing the upper structure and the lower structure in direct contact with each other, wherein the placing vertically aligns the upper structure and the lower structure with each other and a dielectric layer of the upper structure directly contacts a dielectric layer of the lower structure; and performing an annealing process on the upper structure and the lower structure, wherein, after the nitrogen plasma treatment process is performed, copper nitride (Cu.sub.4N) is formed on a top surface of the surface layer and silicon hydroxide (Si—OH) is formed on a top surface of the dielectric layer.
